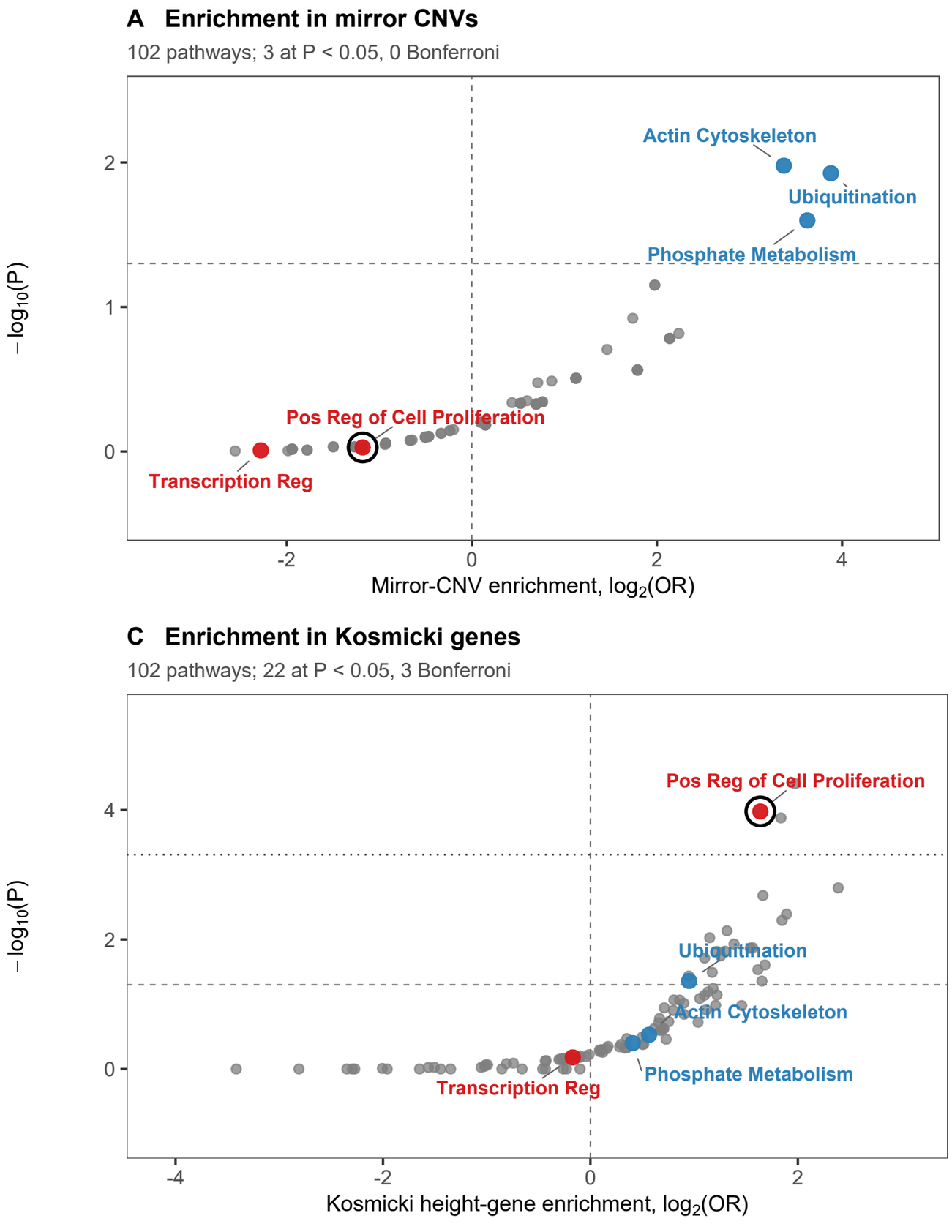
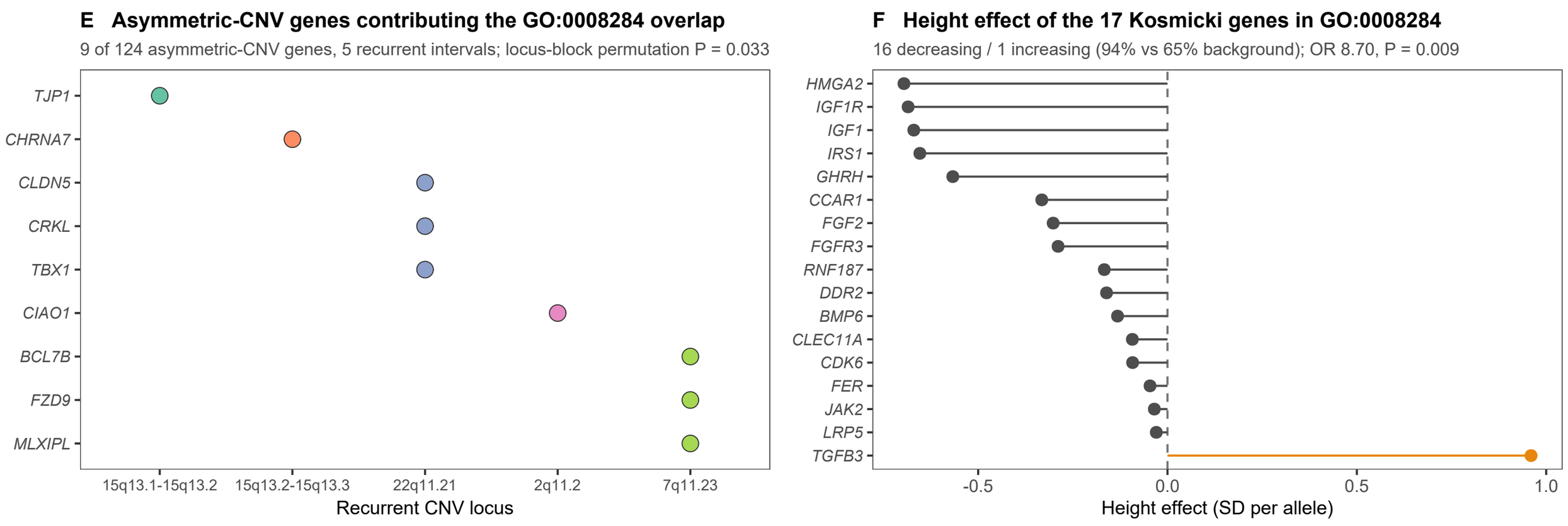
Three-way convergence on a single analysis set: pathway enrichment in mirror (A) and asymmetric (B) CNVs from Sacks et al., pathway enrichment in height genes from Kosmicki et al. (C), and enrichment of height-decreasing effect bias in the pathway's Kosmicki genes relative to all 207 Kosmicki genes (D), with the gene-level evidence behind the convergent pathway GO:0008284 (E, F)

Panels A-D plot the same 102 pathways: the union of those for which the asymmetric CNV test can reach P < 0.05 (>= 6 genes in the CNV gene universe) and those for which the mirror CNV test can (>= 4). That single set fixes the multiple-testing burden throughout, Bonferroni = 0.05/102 = 4.9e-04. In A-D colour marks the three pathways enriched in mirror CNVs (blue) and the two enriched in asymmetric CNVs (red); these groups are disjoint, and only these five carry labels, in the same colours in every panel so each can be followed across the tests. Significance within a panel is read off the threshold lines. The black ring marks the 1 pathway significant in all three tests of the asymmetric analysis (B, C and D): GO:0008284. Panels E and F are gene-level and use their own palettes.





Enriched in mirror CNVs (A) Enriched in asymmetric CNVs (B) Significant in all three tests (B, C, D)

Dashed lines mark OR = 1 and nominal P = 0.05; dotted line marks the Bonferroni threshold where it falls within the plotted range. Neither CNV test shows a screen-level excess (asymmetric 2 hits, mirror 3; chance expects 2.4 for the asymmetric test, which only 48 of the 102 pathways can reach). The contrast between the two CNV classes is a specificity check rather than a formal test. In E, genes within one recurrent locus travel on the same CNV and are not independent, which is why the locus-block permutation, not the gene-level Fisher test, is the statistic of record for the CNV layer.